

Exercise (3.3.3 - Stefano Fochesatto). Prove that if H is a normal subgroup of G of prime index p then for all $K \leq G$ either **(i)** $K \leq H$ or **(ii)** $G = HK$ and $|K : K \cap H| = p$.

Proof: Suppose H is a normal subgroup of G of prime index p and $K \leq G$. Note that since $H \trianglelefteq G$ we know that $N_G(H) = G$ and further that $K \leq N_G(H)$. By the Second Isomorphism Theorem it then follows that $KH/H \cong K/K \cap H$, and $H \leq KH \leq G$. We demonstrated in class that if $H \leq KH \leq G$ then $|G : H| = |G : KH||KH : H|$. Since $KH/H \cong K/K \cap H$ we know that $|KH : H| = |K : K \cap H|$. Therefore by substitution we get,

$$|G : KH||KH : H| = |G : KH||K : K \cap H| = |G : H| = p.$$

(i) Suppose that $|G : KH| = p$ and $|K : K \cap H| = 1$. Therefore for every $k \in K$, $kK \cap H = 1K \cap H$. So $k \in K \cap H$ and therefore $k \in H$. Thus $K \leq H$.

(ii) Suppose that $|G : KH| = 1$ and $|K : K \cap H| = p$. Therefore for every $g \in G$, $gKH = 1KH$. Therefore $g \in KH$ and $G \leq KH$. Since $KH \leq G$ from the Second Isomorphism Theorem, it follows that $G = KH$. Recall that Proposition 14 gives us that $HK = KH$ if either is a subgroup of G . Hence $G = HK$. $\square$